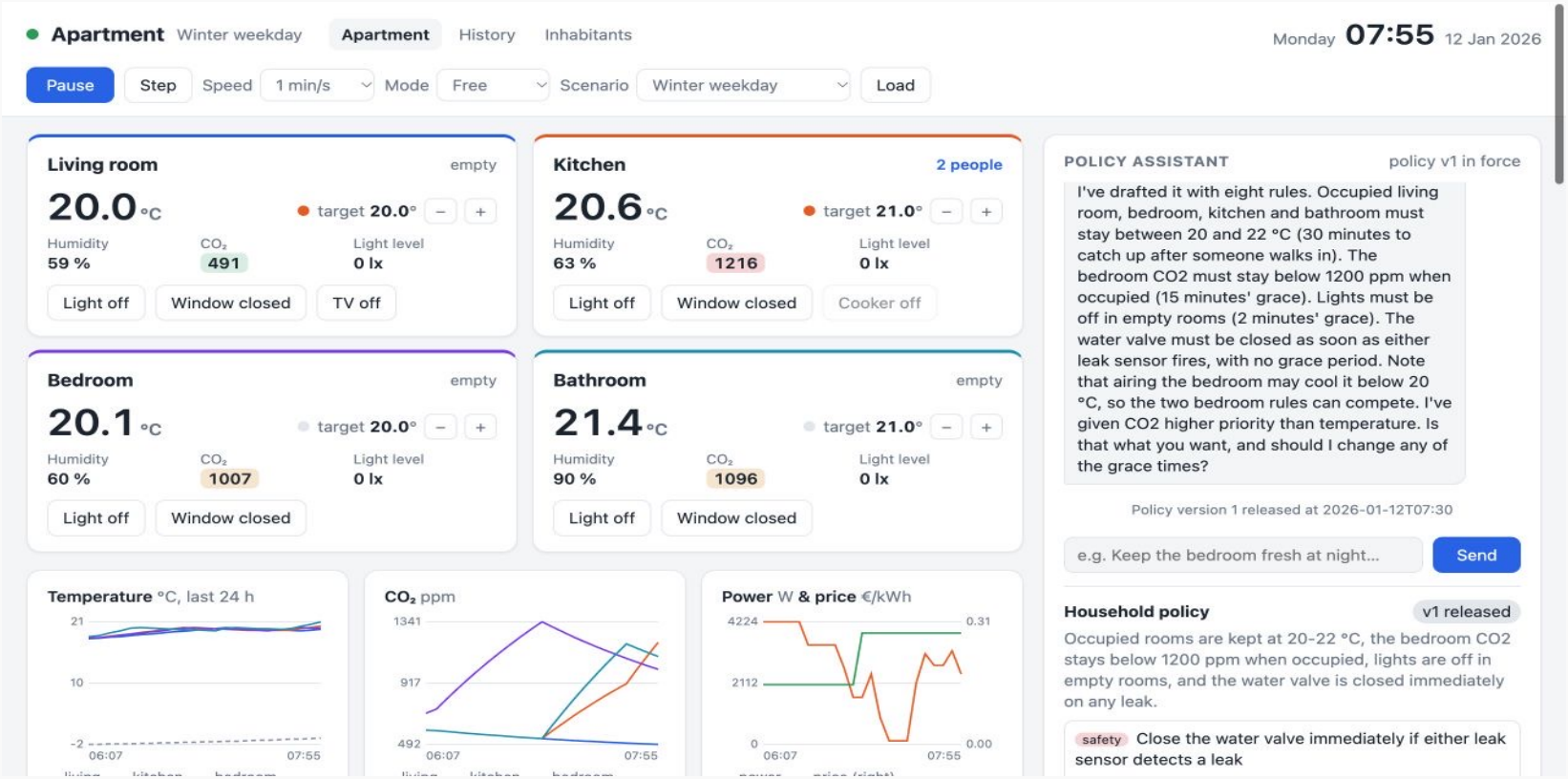
Agentic-driven Policy based control of Smart Homes

Rem Collier
University College Dublin



The apartment demo



LIVE SIMULATOR

Four rooms
WoT devices
Released policy

Temperature, CO_2, occupancy and energy evolve in simulated time.

Winter weekday
1 simulated minute / second

Technology stack

Web Dashboard

Java

Java HTTP Server

HTML

CSS

JavaScript

Simulator

Java

Eclipse Californium

CoAP

Eclipse Paho

Moquette

MQTT

Custom WoT TDs

Agents

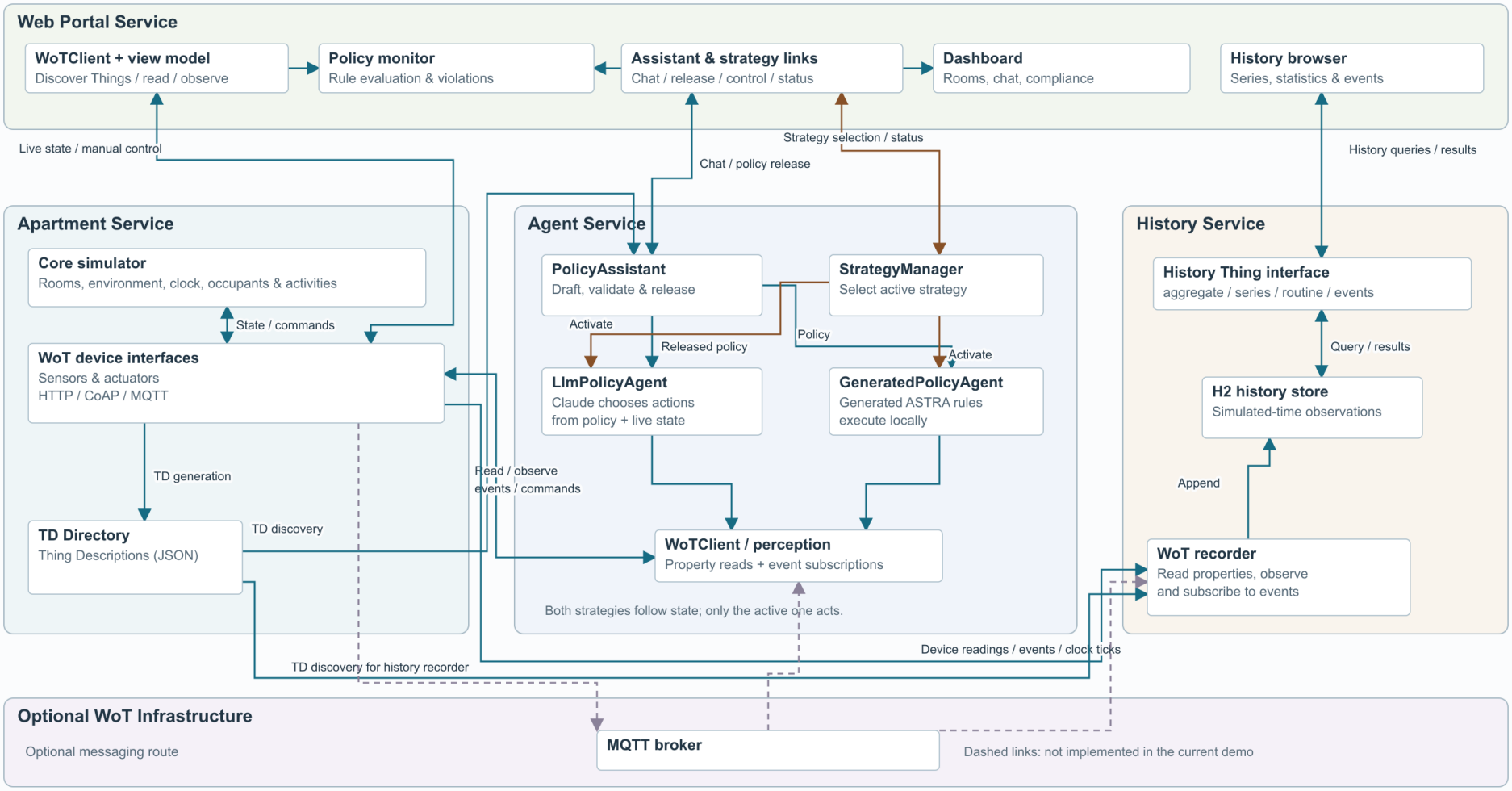
ASTRA/MAMS

astra-langchain4j

mams-astra-jsonschema

mams-astra-wot

Apartment demonstrator — layered service architecture



Solid blue: implemented data / request flow Brown: strategy control Dashed purple: optional broker route • Claude API is external

LLM in the loop: current controller

One decision episode per event or 15-minute simulation tick

1 Build the decision context
Policy + readings + controls + trigger + recent actions

2 Ask Claude
Request JSON actions and a short explanation

Optional history: query → results → ask again (up to 2 rounds)

3 Check proposed operations
Known device / affordance; writable property

4 Execute through WoT
ASTRA writes properties or invokes device actions

Next trigger + updated state

Model response (example)

```
{
 "reasoning":
  "Kitchen is below target.",
 "actions": [
  {
   "thing": "kitchen-thermostat",
   "property": "targetTemperature",
   "value": 21
  }
 ]
}
```

Current design: structured JSON dispatch; no immediate model follow-up after executing device actions.

Generated ASTRA agent strategy

Generate for a policy version; execute locally on each simulation tick

1 Read the released policy
Policy rules + device schemas + current readings

2 Generate ASTRA with Claude
A program for the current policy version

Validation errors → ask Claude to repair (up to 2 attempts)

3 Check, compile and install
Allowed constructs, conditions and device writes

4 Execute every simulated minute
Adopt !enforce(id); issue WoT device commands

Illustrative generated rule

```
plan +!enforce("leak-safety") :
  env.holds("bathroom-leak.leak") {
  !set("water-valve",
      "open", "false");
}
```

When the condition holds, close the water valve.
The rule ID must match the released policy.

Claude is used for generation and repairs; the installed rules then run without model calls.